CareerDrive: Maine Construction Industry Career Map

https://github.com/zongyang078/CareerDrive

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Introduction

Maine's construction and transportation infrastructure industry employs a diverse workforce ranging from entry-level equipment operators to licensed engineers, yet the sector faces persistent labor shortages as retirements outpace new entrants. A key barrier is the lack of a unified platform that helps job seekers understand available career pathways and connect with relevant training programs.

This project addresses that gap. Working with data provided by MaineDOT and AGC Maine, we use unsupervised learning to map 20 key construction occupations, identify natural groupings based on required skills, and quantify the mismatch between training supply and occupational demand. The result is CareerDrive, a Streamlit-based prototype that corresponds to Level 2 of the client's three-phase product roadmap: smart career matching based on a job seeker's background and experience.

Our analysis is structured in three layers:

1. EDA & Supply-Demand Gap — descriptive analysis of the industry ecosystem and training coverage.
2. Unsupervised Learning — Track A (skills-based clustering) and Track B (text clustering + topic modeling), plus network analysis of career pathways.
3. Recommendation Prototype — cosine similarity matching between user-input skills and occupation profiles, delivered as an interactive Streamlit application.

All code, data pipelines, and configuration files needed to reproduce the results in this report are available in the project repository linked on the title page.

Background

Related Work

Career mapping tools such as the Green Buildings Career Map (`https://greenbuildingscareermap.org/`) demonstrate the value of interactive, web-based career pathway visualizations for specialized industries. The O*NET system, maintained by the U.S. Department of Labor, provides standardized occupational descriptors that enable cross-occupation comparison using quantitative features [1].

Unsupervised learning has been applied to labor market analysis in several contexts. K-Means and hierarchical clustering have been used to segment occupations by skill profiles [2], while NMF topic modeling has been applied to job posting text to identify latent skill themes [3]. Network-based approaches using centrality metrics and community detection have been used to model occupational mobility and skill transferability [4].

O*NET Data

O*NET 30.2 provides importance and level ratings for 35 skills, 33 knowledge areas, and 52 abilities per occupation, yielding a 120-dimensional feature vector. These features serve as the basis for all quantitative analyses in this report. Text enrichment was obtained via the O*NET Web Services API v2, which provides free academic access with registration.

Methods and Analysis

Data Sources and Collection

Our analysis integrates three data sources, summarized in Table 1.

Table 1: Data Sources

Source	Records	Description
O*NET 30.2 Database	20 × 120	Skills, knowledge, abilities
O*NET API v2	20 occupations	Tasks, work activities, technology (text)
Client Excel	8 sheets	Companies, apprenticeships, education programs

Starting from 28 occupation rows in the client Excel, we extracted 21 unique O*NET codes, of which 20 had sufficient data in the local O*NET 30.2 database (code 13-1082.00 was excluded due to missing feature data). Company data was merged from the AGC Maine member list (222 companies) and the MaineDOT prequalified contractor list (57 companies), yielding 272 unique firms after normalizing and deduplicating by company name.

Data Cleaning

Feature matrix. The 20 × 120 feature matrix was constructed from O*NET importance ratings (scale 1–5). No missing values were present after filtering to valid occupation codes.

Education program data. The Community College Programs sheet in the client Excel contained mixed content: 14 actual course records followed by non-course rows describing funding programs, immersion initiatives, and lab facilities with a different column schema. We identified and retained only rows corresponding to known Maine community college codes (CMCC, EMCC, KVCC, NMCC, SMCC, WCCC, YCCC), reducing the dataset from 21 to 14 valid records.

Occupation-cluster mappings. Manual mappings linking apprenticeship titles, CC programs, and UMaine programs to O*NET codes or cluster labels were initially inconsistent: the CC mapping contained 15 entries referencing program names that did not exist in the cleaned data, and the UMaine mapping referenced non-existent programs such as "Electrical Engineering Technology" (actual name: "Electrical & Computer Engineering"). After correcting all names to match actual records, matched CC programs increased from 5/21 to 12/14 and UMaine programs from 6/10 to 8/10. All mappings are now centrally managed in `data/mappings.json` to ensure reproducibility.

Exploratory Data Analysis

The 272 companies span five types: Specialty Contractors (67), General Contractors (60), Service Providers (58), DOT Prequalified (50), Suppliers (35), and Developer (2). Specialty and general contractors together represent the majority of the ecosystem, consistent with Maine's project-based construction economy.

Supply-demand gap analysis was conducted by comparing the number of occupations in each cluster (demand) against available training pathways — apprenticeships plus education programs (supply). As shown in Figure 1, the Entry Level/Operators cluster faces the largest gap, with only one apprenticeship pathway mapped to its ten occupations. Figure 2 and Figure 3 show the breakdown of apprenticeship and education coverage by cluster respectively.

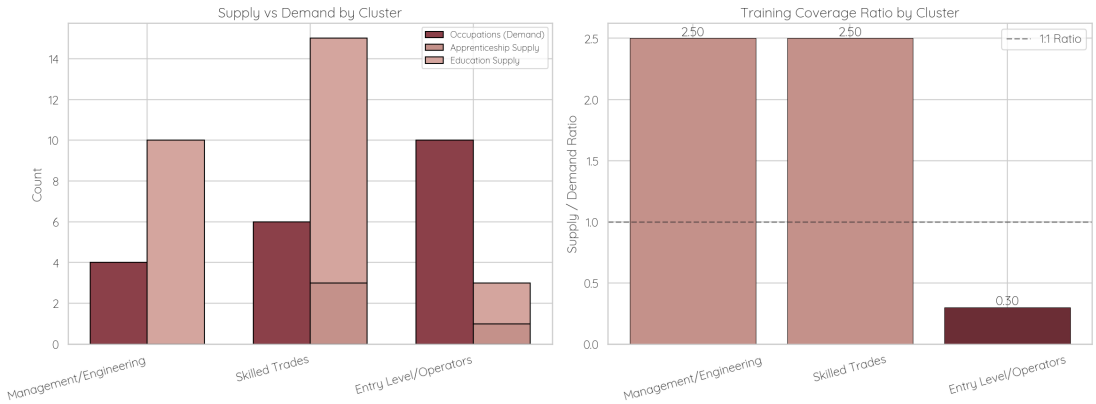

Figure 1: Supply-demand gap by cluster. Dark bars (right panel) indicate clusters where Coverage Ratio < 1.0. Entry Level/Operators has the most severe training shortfall.

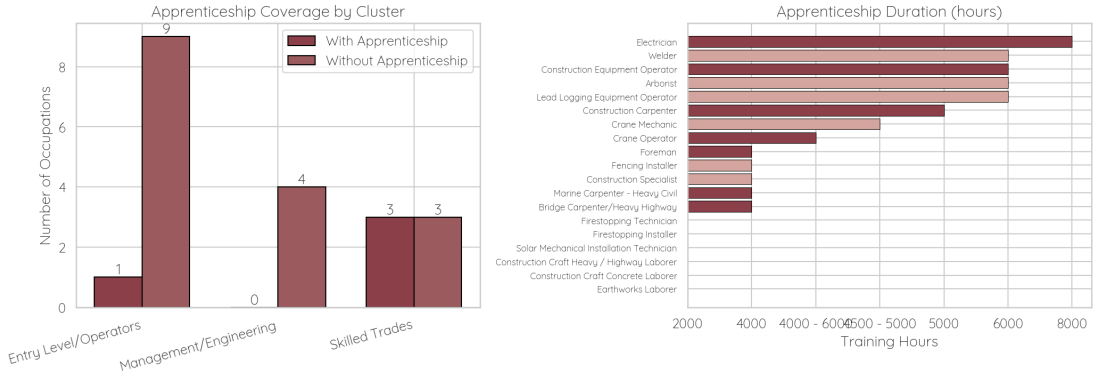

Figure 2: Apprenticeship pathway coverage by cluster and occupation. Only 7 of 19 apprenticeships map to the 20 occupations in scope.

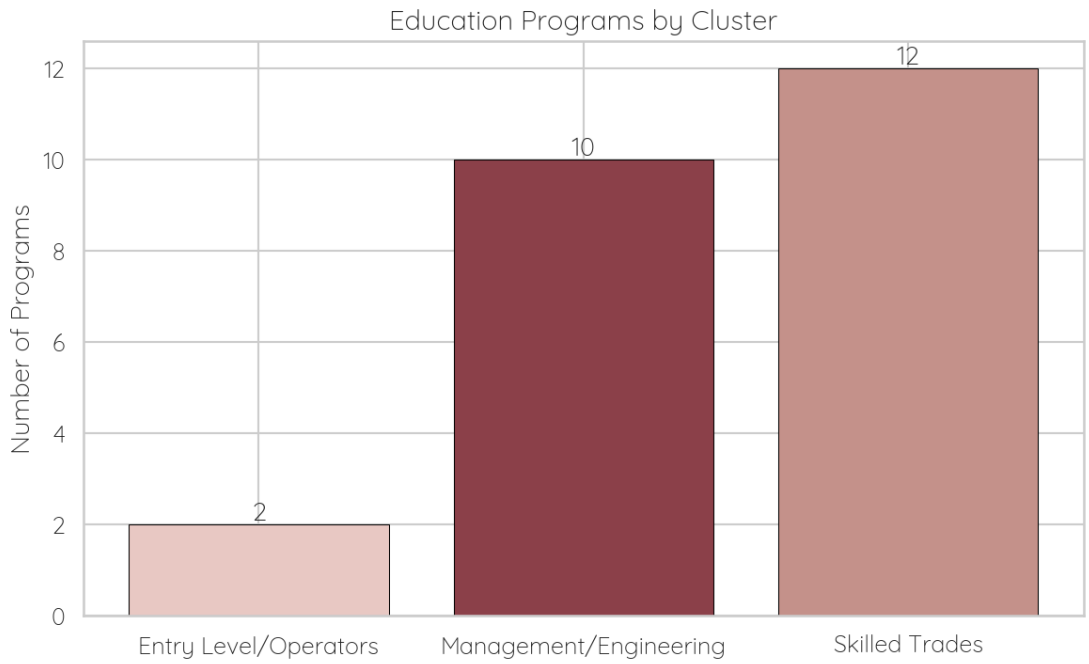

Figure 3: Education program coverage by cluster after mapping correction (CC: 12/14 matched; UMaine: 8/10 matched).

Track A: Skills-Based Clustering

Preprocessing. The 120-dimensional feature matrix was standardized using `StandardScaler` to ensure equal weighting across features with different natural scales.

Dimensionality reduction. PCA was applied to the scaled matrix. The first principal component (PC1: Cognitive & Communication Complexity) explains 53.6% of total variance, capturing oral expression, reading comprehension, time management, and mathematical reasoning — skills that distinguish higher-complexity, knowledge-intensive occupations from operational roles. The second component (PC2: Perceptual & Sensory Coordination) explains 9.0% of variance and is dominated by visual color discrimination, near vision, time-sharing, and auditory attention — abilities that characterize hands-on operational roles. Together, PC1 and PC2 capture 62.7% of total variance and reflect high inherent structure in the occupation feature space.

Clustering. K-Means clustering was applied with $k = 3$, selected via elbow method and silhouette analysis. The resulting silhouette score of 0.315 indicates moderate but meaningful cluster separation, appropriate given the small sample ($n = 20$) and high-dimensional input. DBSCAN was evaluated as an alternative but proved unsuitable due

to the uniform density structure of the feature space at this scale.

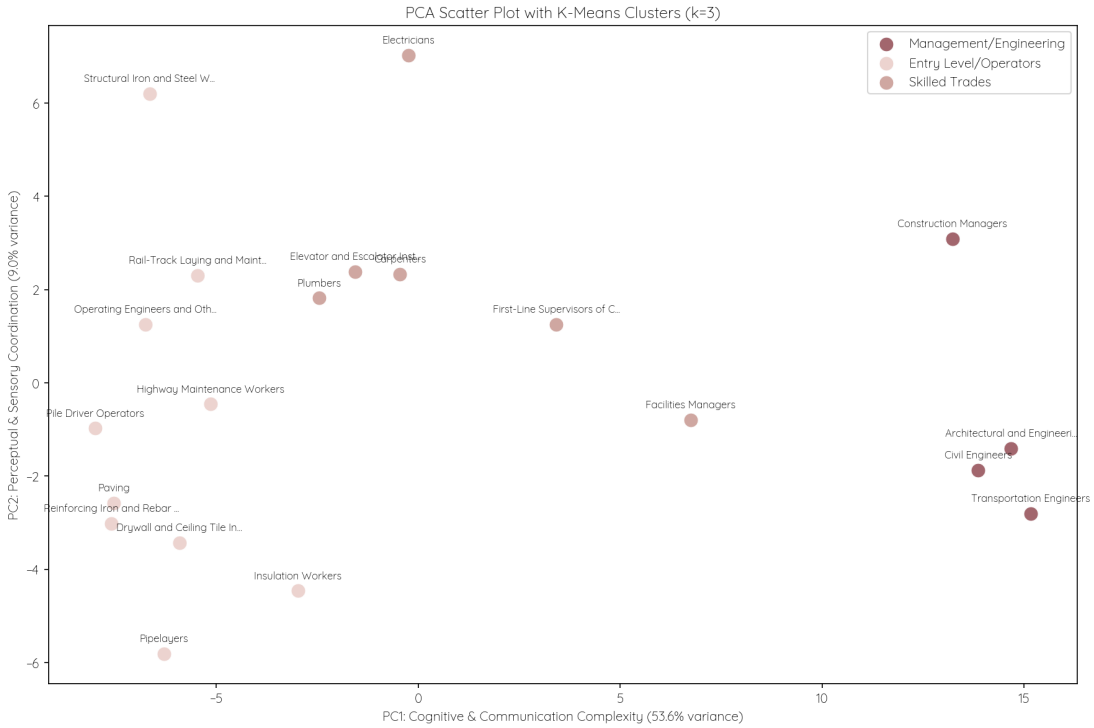

Figure 4: PCA scatter plot with K-Means cluster assignments ($k = 3$). The horizontal axis (PC1: Cognitive & Communication Complexity, 53.6%) separates Management/Engineering from Entry Level clusters. The vertical axis (PC2: Perceptual & Sensory Coordination, 9.0%) distinguishes operational and hands-on roles from knowledge-intensive ones.

The three clusters shown in Figure 4 correspond to meaningful occupational groupings summarized in Table 2. Alignment with O*NET Job Zones provides external validation: Management/Engineering occupations require bachelor's degrees or higher (Zones 4–5), while Entry Level/Operators require only short-term training (Zone 2).

Table 2: Cluster Composition and Job Zone Alignment

Cluster	Occupations	Job Zones
Management / Engineering	4	4–5
Skilled Trades	6	2–3
Entry Level / Operators	10	2
Overall silhouette score: 0.315		

Figure 5 shows the top distinguishing features for each cluster. Management/Engineering

occupations score highest on knowledge of Engineering and Technology, while Entry Level/Operators are characterized by equipment operation and monitoring skills.

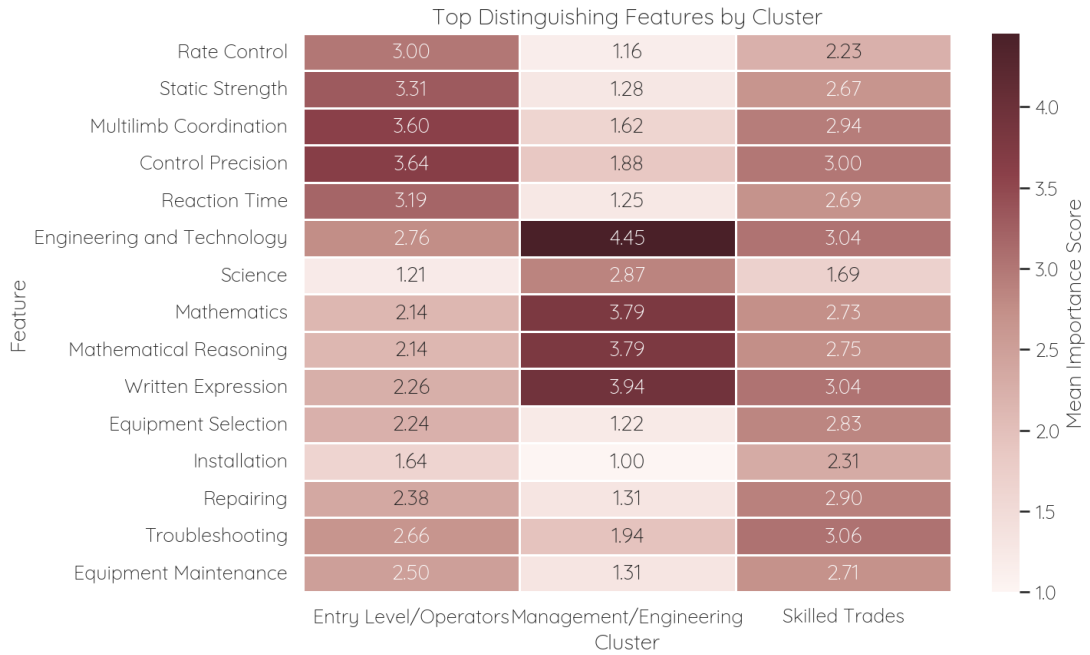

Figure 5: Heatmap of top distinguishing O*NET features by cluster. Values represent mean importance scores (scale 1–5).

Track B: Text Clustering and Topic Modeling

Text collection. Occupation descriptions were enriched using the O*NET API v2, combining tasks, work activities, and technology fields into a single text document per occupation (average 1,417 characters).

Vectorization. TF-IDF vectorization [6] with 200 features, unigram and bigram tokens, and `min_df=2` was applied to the 20-document corpus.

K-Means on text features. K-Means clustering on TF-IDF vectors yielded a silhouette score of 0.103 — lower than Track A but expected given that 20 documents is a very small corpus for text-based clustering.

NMF topic modeling. Non-negative Matrix Factorization (NMF) with 3 components extracted three interpretable topics, shown in Table 3 and visualized in Figure 6.

Table 3: NMF Topics Extracted from O*NET Occupation Text

Topic	Label	Top Keywords
0	Equipment Operations	equipment, machine, operate, software
1	Engineering & Management	construction, design, project, engineering
2	Installation & Fabrication	steel, install, blueprints, pipe

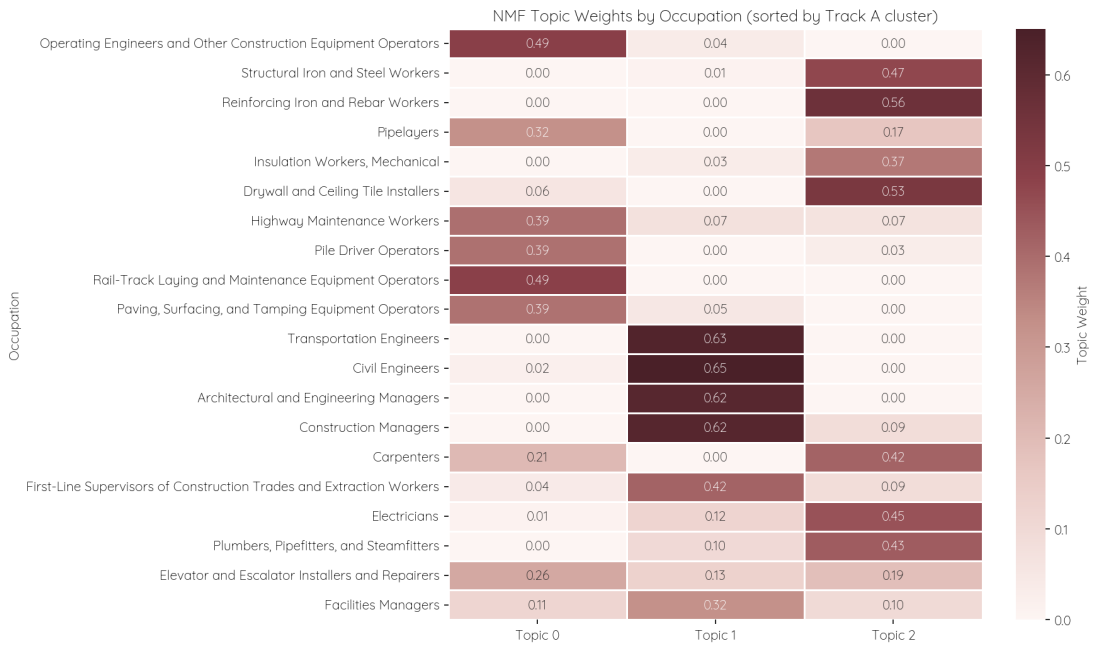

Figure 6: NMF topic weight heatmap by occupation, sorted by Track A cluster assignment. Management/Engineering occupations load strongly on Topic 1; Entry Level/Operators on Topic 0.

Cross-validation with Track A shows that the Management/Engineering cluster achieves 4/4 perfect topic alignment. Overall agreement between NMF topic assignment and K-Means cluster is 65% (Adjusted Rand Index = 0.269), reflecting meaningful but imperfect correspondence — as expected when comparing skill-profile clustering with text-derived topics.

Network Analysis

Graph construction. A career path network was constructed with 20 occupation nodes. Edges were drawn from two sources: (1) O*NET Related Occupations relationships (51 edges, representing officially recognized career pathways), and (2) cosine similarity

edges where pairwise similarity of scaled feature vectors exceeded 0.95. The resulting graph has density = 0.268 and forms a single connected component.

Centrality metrics. Three centrality measures were computed to identify structurally important occupations, summarized in Table 4.

Table 4: Top Occupations by Network Centrality Metric

Metric	Occupation	Value
Degree (hub)	Structural Iron & Steel Workers	10
Betweenness (bridge)	Electricians	0.244
PageRank	Structural Iron & Steel Workers	0.092

Structural Iron & Steel Workers functions as a network hub — connected to the most other occupations — suggesting it is a pivotal role within the Maine construction ecosystem. Electricians, despite lower degree, have the highest betweenness centrality, indicating they serve as a critical bridge between clusters and represent a key skill transfer pathway.

Community detection. Louvain community detection [5] identified 3 communities (ARI = 0.279 vs. K-Means clusters), with 100% agreement on the Management/Engineering grouping. The moderate overall ARI reflects that network topology and skill-profile similarity capture related but distinct aspects of occupational structure.

Results

Occupational Clustering

All three analytical methods independently identify the same three-group structure among the 20 Maine construction occupations, as summarized in Table 5.

Table 5: Cross-Method Convergence on Three-Group Structure

Method	Groups	Mgmt/Eng Match	ARI vs. K-Means
K-Means (Track A)	3	Reference	—
NMF Topics (Track B)	3	4/4 (100%)	0.269
Louvain (Network)	3	4/4 (100%)	0.279

The convergence is notable: three methods operating on fundamentally different representations — numeric skill profiles, unstructured occupation text, and a relational career path graph — all recover the same Management/Engineering, Skilled Trades, and Entry Level/Operators partition. The Management/Engineering cluster achieves 100% agreement across all three methods, providing the strongest structural signal. The moderate ARI values (0.269–0.279) for the remaining two clusters are expected: NMF and Louvain capture textual and topological similarity respectively, which partially diverge from pure skill-profile distance at the Skilled Trades and Entry Level boundary.

External validation against O*NET Job Zones confirms the clustering is meaningful: Management/Engineering occupations (Job Zones 4–5) require at least a bachelor's degree, while Entry Level/Operators (Job Zone 2) require only short-term on-the-job training. The K-Means silhouette score of 0.315 indicates moderate separation, appropriate for a dataset of only 20 occupations.

Network Structure

Within the career path network, Structural Iron & Steel Workers (degree = 10, PageRank = 0.092) is the most connected occupation and functions as the ecosystem hub. Electricians, despite having lower degree, hold the highest betweenness centrality (0.244), indicating they serve as the primary bridge between the Skilled Trades and Management/Engineering clusters. This suggests Electricians represent a high-leverage upward mobility pathway in the Maine construction workforce.

Supply-Demand Gap

The supply-demand gap analysis reveals a clear training shortfall concentrated in the Entry Level/Operators cluster. With 10 occupations but only 1 mapped apprenticeship pathway, this cluster has the lowest coverage ratio of the three. By contrast, Skilled Trades benefits from 12 of 14 community college programs correctly mapped after data correction, and Management/Engineering is relatively well served by UMaine engineering degree programs (8 of 10 matched). Table 6 summarizes coverage by cluster.

Table 6: Training Coverage by Cluster

Cluster	Occupations	Apprenticeships	CC / UMaine Programs
Management / Engineering	4	2	8 UMaine
Skilled Trades	6	4	12 CC
Entry Level / Operators	10	1	0 CC / UMaine

Recommendation Prototype

The CareerDrive Streamlit application operationalizes the clustering and network results into three user-facing pages. The Career Match page accepts a job seeker's plain-language background description across seven skill categories and returns ranked occupation matches via cosine similarity, together with cluster membership and available training pathways. This constitutes a working prototype of the Level 2 product envisioned in the client roadmap.

Discussion

What the Results Mean

The convergence of three independent methods on the same three-cluster structure provides confidence that these groupings reflect genuine occupational differences rather than artifacts of any single algorithm or data representation. The alignment with O*NET Job Zones further supports external validity.

The supply-demand gap finding has direct policy implications: resources for apprenticeship and training program development should prioritize the Entry Level/Operators cluster, which includes roles such as Pipelayers, Drywall Installers, and Highway Maintenance Workers — occupations with low barriers to entry but insufficient structured training pathways in Maine.

How Results Can Be Used

The CareerDrive Streamlit prototype demonstrates three concrete use cases for MaineDOT:

1. Industry Dashboard: Gives MaineDOT staff a view of the company ecosystem and training coverage gaps by cluster.
2. Career Path Map: An interactive force-directed network visualization showing how occupations relate by skill similarity and official O*NET pathways, supporting workforce planning discussions.
3. Career Match: Allows job seekers to describe their background in plain language and receive occupation recommendations with training pathway information — a prototype of the Level 2 product envisioned by the client.

Limitations

1. Small sample size. With only 20 occupations, clustering results are sensitive to random seed initialization. The silhouette score of 0.315 indicates moderate separation; conclusions should be interpreted with caution and validated as the occupation scope expands.
2. Supply-side metric is approximate. Coverage Ratio treats all programs equally regardless of enrollment capacity. A 5-person certificate program and a 200-seat AAS degree count equally in our current analysis.
3. Data quality required manual correction. The CC program data in the client Excel mixed course rows with non-course content (funding initiatives, lab descriptions). After filtering to known college codes, valid records decreased from 21 to 14. Mapping corrections increased matched programs from 5/21 to 12/14.
4. Apprenticeship scope limitation. 7 of 19 apprenticeship programs map to our 20 defined occupations. The remaining 12 target roles (e.g., Welders, Plumbers) fall outside our defined occupation scope — this is a research scope decision, not a data gap.
5. Static data. All company and program records reflect a single point in time. The prototype does not integrate live job listings or real-time enrollment data.

Future Work

Based on the client's three-phase roadmap and our findings, we recommend the following next steps:

1. Level 2 full build. Integrate real-time job listing APIs (Indeed, LinkedIn) and live program enrollment data to replace the current static snapshot.
2. Level 3: conversational AI career coach. The client roadmap envisions a small language model (SLM) trained on DOT-specific data. Our occupation clusters and career path network could serve as structured knowledge for retrieval-augmented generation (RAG).
3. Expand occupational scope. Extend from 20 construction occupations to the full Maine transportation infrastructure sector, including aviation, maritime, and urban planning roles.

4. Geographic dimension. Incorporate county-level employment and wage data from BLS to enable regional supply-demand analysis within Maine.
5. Feedback loop. Log Career Match usage patterns to refine the recommendation model over time, moving from cosine similarity toward a learned ranking model.

References

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